

Sales Forecasting in FMCG Using Machine Learning

Introduction



Objective: Forecast future sales volume using ML models.



Importance: Inventory planning, supply chain, and marketing.



Data includes: price, promotions, category, cost, region, day.

Problem Statement

Predict sales using:

Product
characteristics
(price, promotions,
category, cost)

Store info (region,
day)

Challenges:

Timeseries trends

Multicollinearity

Outliers

Missing data
(handled)

Dataset Overview

Industry Focus: Fast-Moving Consumer Goods (FMCG).

Objective: To understand sales dynamics and forecast future sales.

Size:

1000 rows and 9 features.

Columns:

Independent: Date, new_price, Product_Category,
Store_Location, Promotion, Weekday,
Replenishment_Lead_Time, Supplier_Cost

Dependent: Sales_Volume

•**Process Steps:**

Conducted **Exploratory Data Analysis (EDA)**.

Applied **data preprocessing** techniques.

Trained **multiple model** to predict Sales_Volume.

Testing:

Used a **synthetic test dataset** generated by ChatGPT for validation.



```
[ ] # Display the first few rows of the dataframe  
df.head()
```



	Date	new_price	Product_Category	Store_Location	Promotion	Weekday	Replenishment_Lead_Time	Supplier_Cost	Sales_Volume
0	1/1/2022	9.347633	0	0	0	5	9	9.299281	471
1	2/1/2022	16.673568	1	0	0	6	5	13.274109	527
2	3/1/2022	15.823768	2	1	0	0	9	13.302265	581
3	4/1/2022	16.968596	1	0	1	1	5	10.056158	530
4	5/1/2022	4.309673	1	1	1	2	8	3.562862	880

Let's check the data types and convert the 'Date' column to a datetime object.

```
[ ] # Display basic descriptive statistics
    print(df.describe())
```



	new_price	Product_Category	Store_Location	Promotion \
count	1000.000000	1000.000000	1000.000000	1000.000000
mean	12.827163	1.994000	0.978000	0.491000
std	4.592544	1.435278	0.813743	0.500169
min	1.300601	0.000000	0.000000	0.000000
25%	9.520554	1.000000	0.000000	0.000000
50%	13.226681	2.000000	1.000000	0.000000
75%	16.347590	3.000000	2.000000	1.000000
max	27.541569	4.000000	2.000000	1.000000

	Weekday	Replenishment_Lead_Time	Supplier_Cost	Sales_Volume
count	1000.000000	1000.000000	1000.000000	1000.000000
mean	2.999000	4.970000	7.552047	793.268000
std	2.001751	2.589944	4.163729	292.442279
min	0.000000	1.000000	0.500445	0.000000
25%	1.000000	3.000000	3.887763	580.750000
50%	3.000000	5.000000	7.422058	797.500000
75%	5.000000	7.000000	11.099853	1001.250000
max	6.000000	9.000000	14.993587	1507.000000

Data Preprocessing

Steps taken:

Handled missing
values

Feature scaling for
linear models only

Feature Scaling

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
df[['new_price', 'Supplier_Cost', 'Sales_Volume']] = scaler.fit_transform(df[['new_price', 'Supplier_Cost', 'Sales_Volume']])
```

```
[ ] print("Shape of data: ",df.shape) # checking for the shape of
```

➡ Shape of data: (1000, 9)

```
[ ] # Check data types
df.info()
```

➡

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Date                 1000 non-null   object
1   new_price            1000 non-null   float64
2   Product_Category    1000 non-null   int64
3   Store_Location       1000 non-null   int64
4   Promotion            1000 non-null   int64
5   Weekday              1000 non-null   int64
6   Replenishment_Lead_Time 1000 non-null   int64
7   Supplier_Cost        1000 non-null   float64
8   Sales_Volume         1000 non-null   int64
dtypes: float64(2), int64(6), object(1)
memory usage: 70.4+ KB
```



```
# Summary statistics  
df.describe()
```



	new_price	Product_Category	Store_Location	Promotion	Weekday	Replenishment_Lead_Time	Supplier_Cost	Sales_Volume
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	12.827163	1.994000	0.978000	0.491000	2.999000	4.970000	7.552047	793.268000
std	4.592544	1.435278	0.813743	0.500169	2.001751	2.589944	4.163729	292.442279
min	1.300601	0.000000	0.000000	0.000000	0.000000	1.000000	0.500445	0.000000
25%	9.520554	1.000000	0.000000	0.000000	1.000000	3.000000	3.887763	580.750000
50%	13.226681	2.000000	1.000000	0.000000	3.000000	5.000000	7.422058	797.500000
75%	16.347590	3.000000	2.000000	1.000000	5.000000	7.000000	11.099853	1001.250000
max	27.541569	4.000000	2.000000	1.000000	6.000000	9.000000	14.993587	1507.000000

Exploratory Data Analysis (EDA)



SALES VARY BY
REGION AND
CATEGORY.



PROMOTIONS
BOOST SALES.



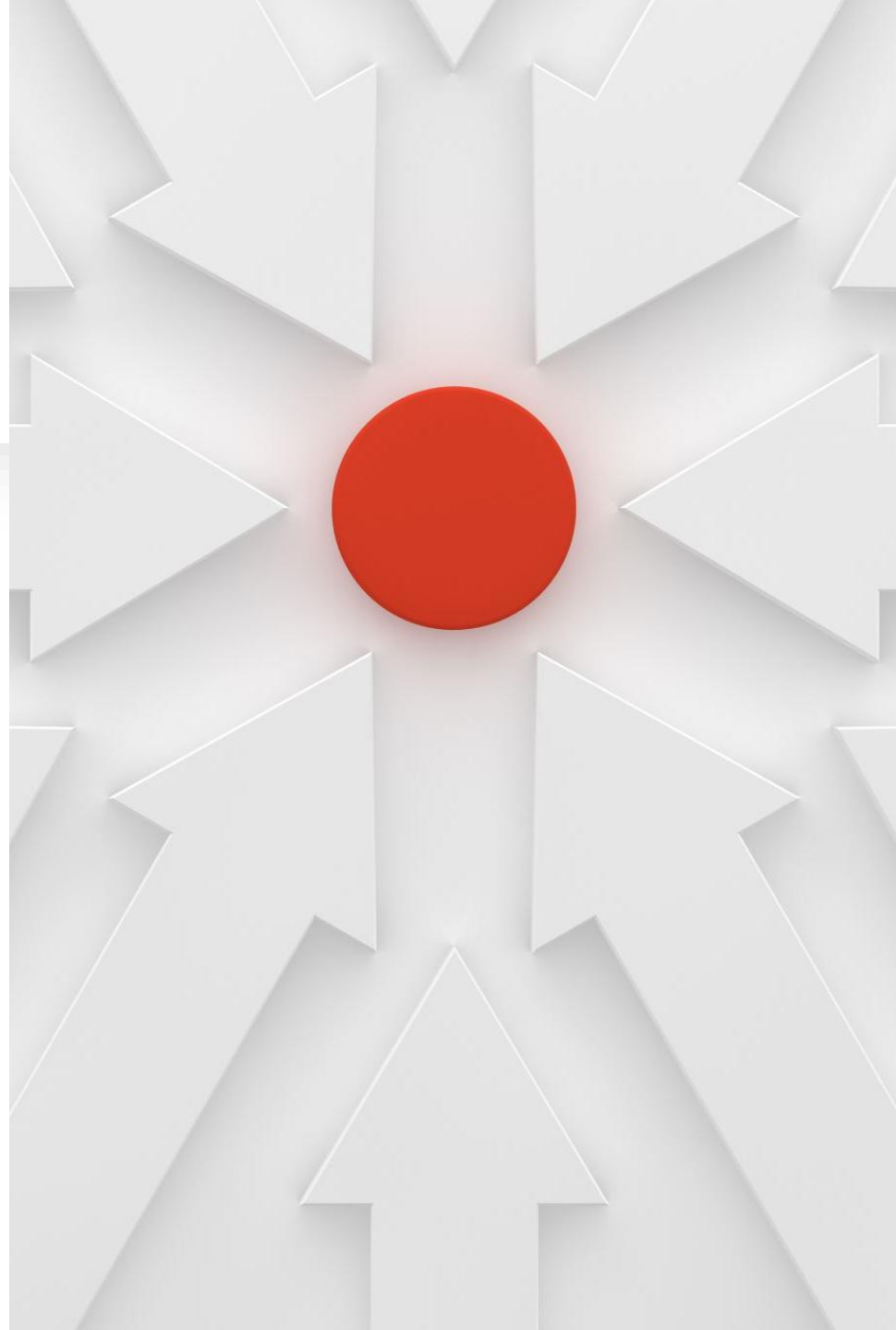
NEW_PRICE AND
SUPPLIER_COST
CORRELATED.



DATA CLEAN AND
READY.

Key Visualizations

- Price vs Sales: No strong trend
- Outliers in sales volume
- Time Series shows volatility
- Weak feature correlation



Plot Overview: The scatter plot shows the relationship between product price and sales volume.

X-Axis: Represents the new price of the product.

Y-Axis: Represents the corresponding sales volume of the product.

Data Distribution: The data points are evenly distributed.

Observation: No clear correlation between higher or lower prices and sales volumes.



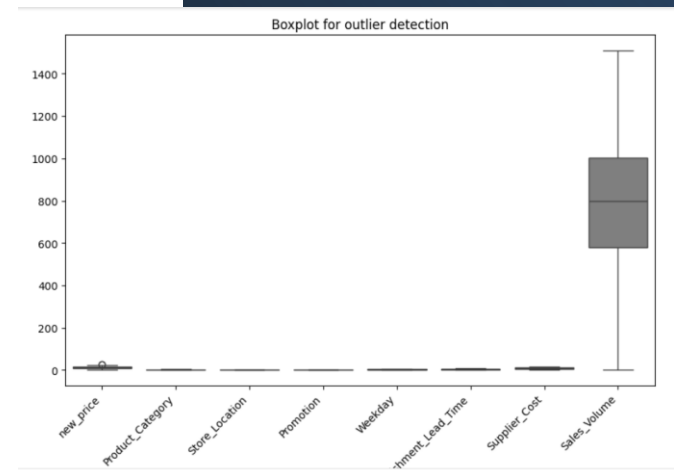
Outlier detection

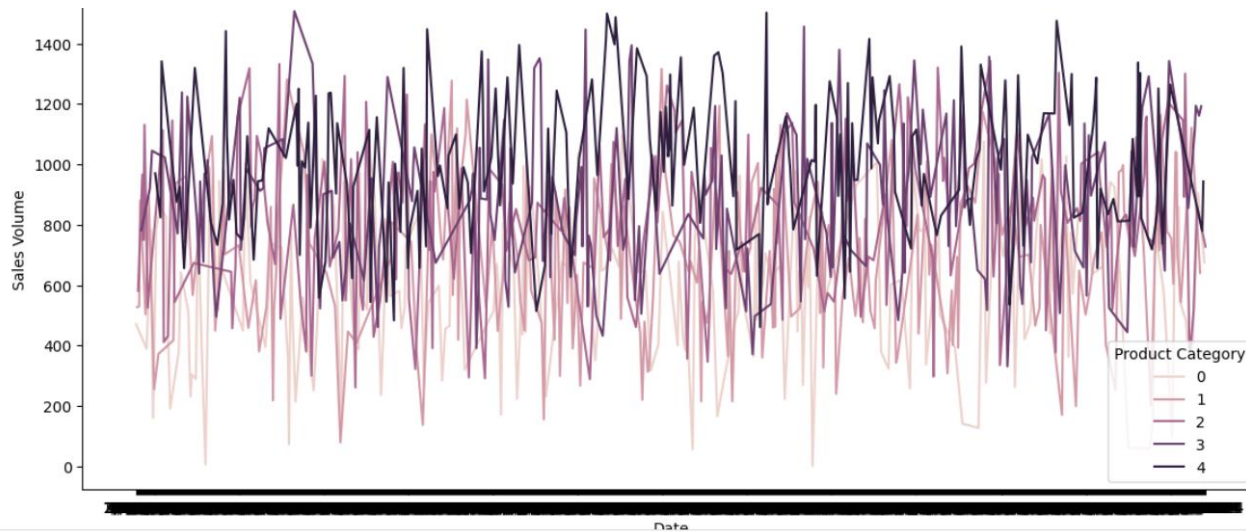
Plot Type: Boxplot illustrating distribution of sales volume across variables.

Overall Insight: Most variables show limited influence on sales volume.

Sales Volume Distribution: Sales volume displays some variation.

Outliers: There are no outliers as there are no datapoints lying outside the whiskers.



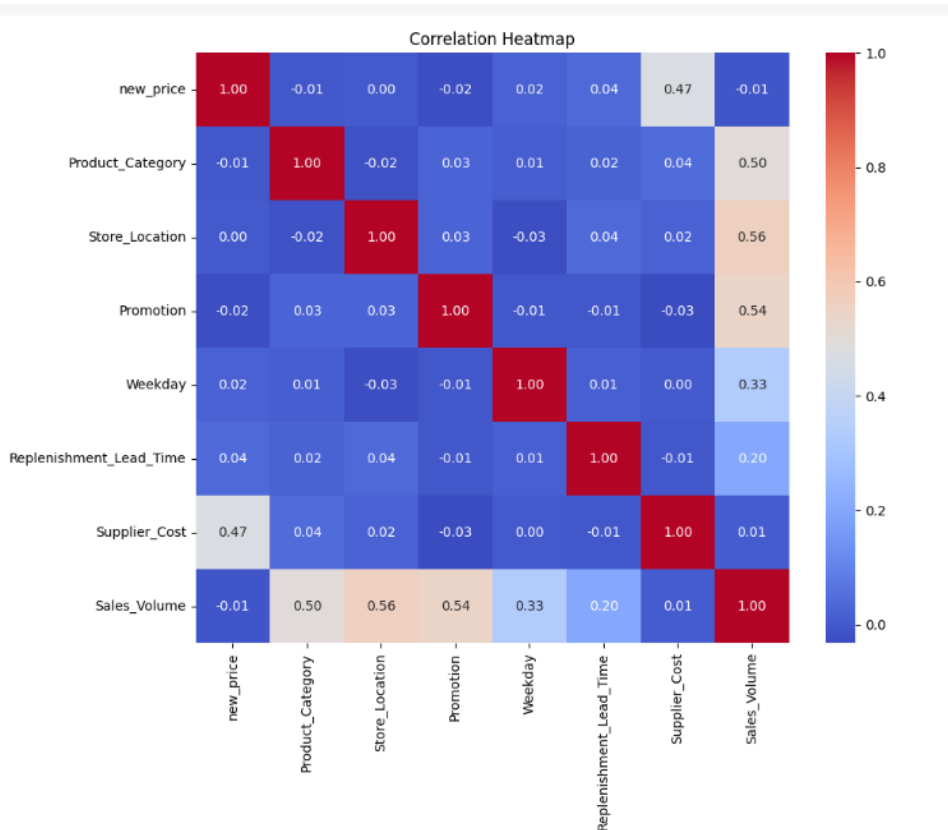


Sales Volatility: Sales volume fluctuates significantly over time.

Demand Instability: Indicates inconsistent demand patterns.

Possible Influencing Factors: Changes may be due to promotions, holidays, or product launches.

Sales Volume Over Time



Correlation Heatmap

Sales Correlation: Sales volume shows a strong relationship with certain variables.

Store Location: Sales vary depending on geographic or specific store locations.

Product Category: Different categories impact sales volume differently.

Promotions: Promotional activities positively influence sales volume.

Model Building Overview

Multiple
models are
used

Linear
Regression

Random
Forest

Decision Tree

Ridge & Lasso
Regression

Prophet

Linear Regression

MSE: 925

R^2 : 0.98

Strong fit

But overfitting?

Other Models

Random Forest:

$R^2=0.97$,
MSE=1998

Decision Tree:

$R^2=0.85$,
MSE=14748

Ridge:

$R^2=0.98$,
MSE=1161

Lasso:

MSE: 943
R-squared: 0.98

Overfitting Check

Dataset Creation: A synthetic dataset for 6 months in 2025 was generated using ChatGPT.

Model Evaluation: Linear regression model was trained and tested on this new dataset.

Performance Metrics: Training R^2 and Testing R^2 are both approximately **0.99**.

Overfitting Check: Close R^2 values suggest **no overfitting**.

Conclusion: Linear regression model is performing well and generalizes effectively. The model is not overfitted as we have used a test data and multiple models.

Prophet Forecasting

Used for trend visualization

Graph includes:

Dots: Actual

Line: Forecast

Shade: Confidence interval

Prophet
Trend
Analysis

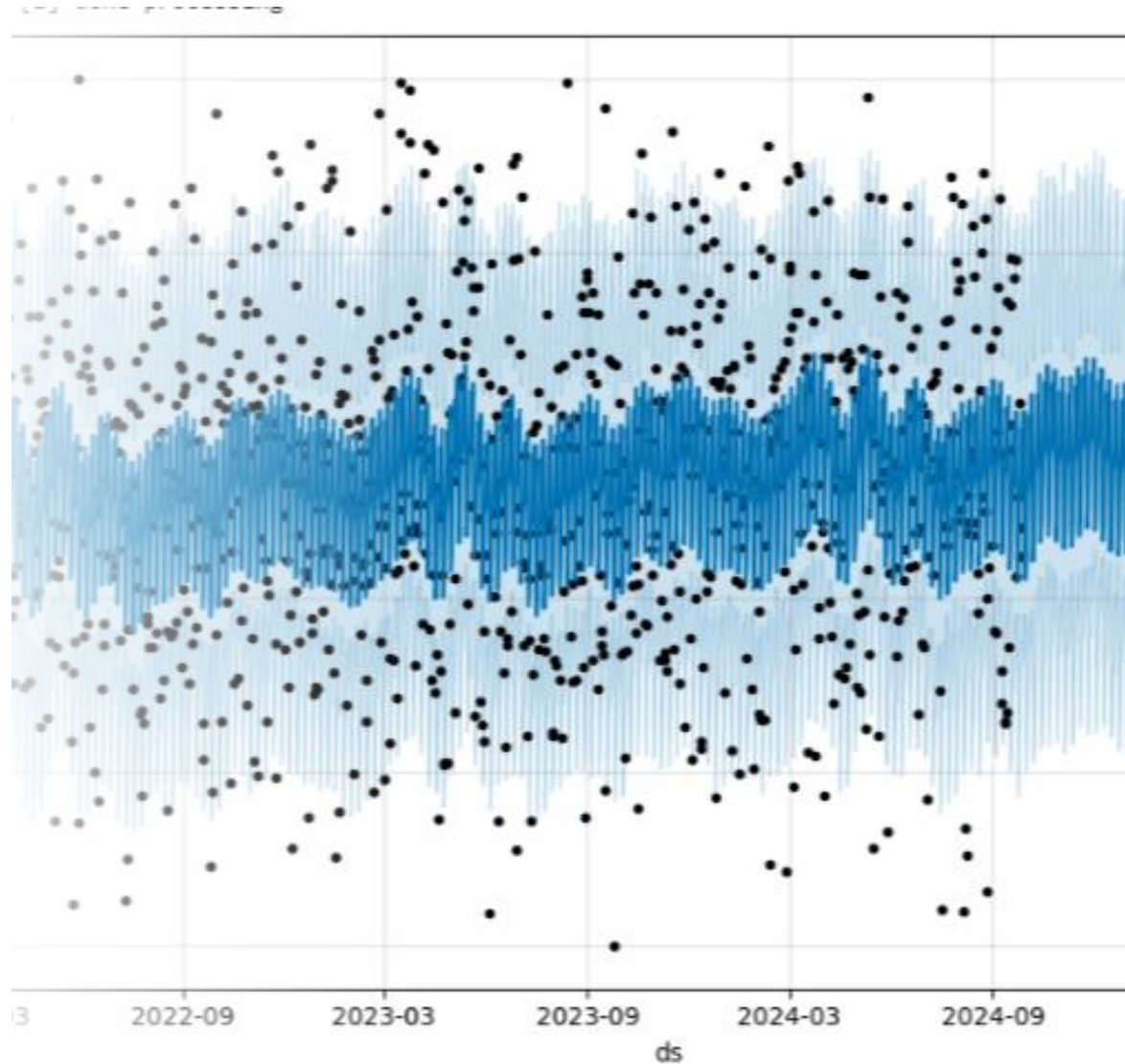
Has four
visualization

Shows
seasonality

Gives Business
Insights

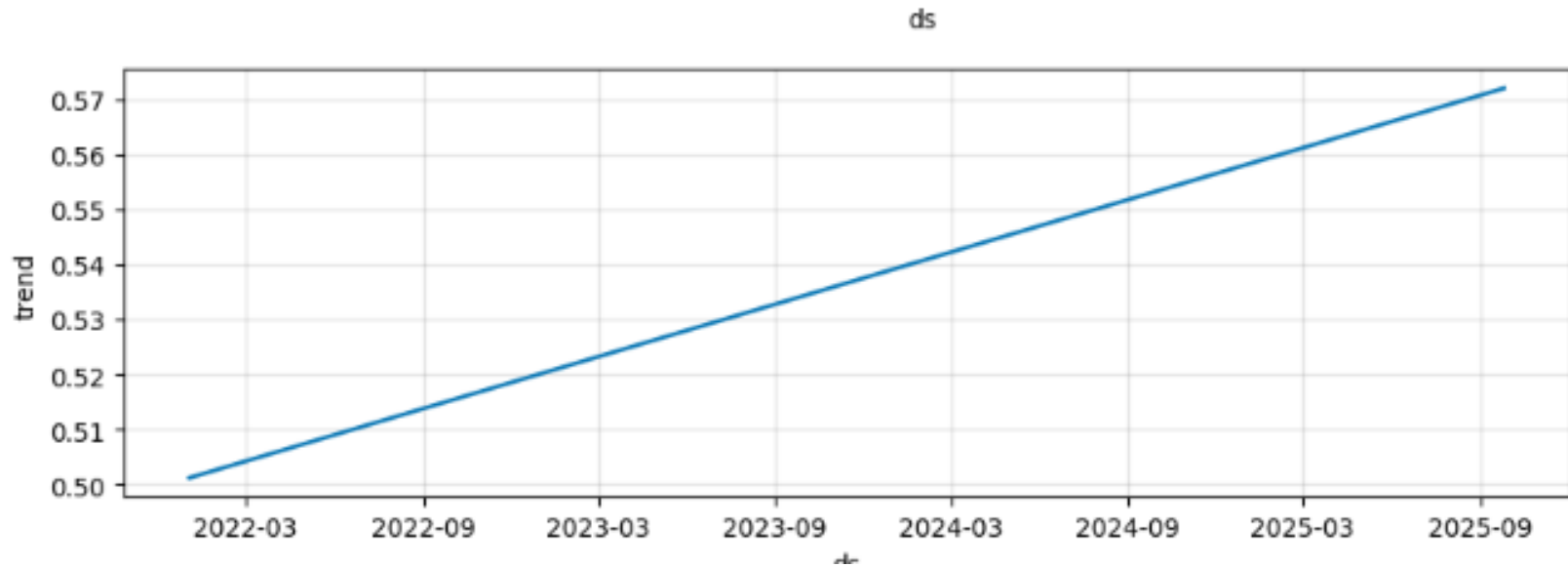
Time series forecast

- **Model Used:** Forecast generated using Prophet time series model.
- **Time Frame:** Covers data from early 2022 to projected values into late 2025.
- **Forecast Accuracy:**
 - Historical data (black dots) closely follows the forecast line, indicating a **good model fit**.
- **Uncertainty:**
 - The **light blue shaded area** represents prediction intervals, which **widen over time**, showing increasing uncertainty.
- **Data Characteristics:**
 - Actual data points show **high volatility**, while the forecast shows a **smoothed trend**.
- **Future Prediction:**
 - The model expects sales to **remain stable**, with no major upward or downward trends.



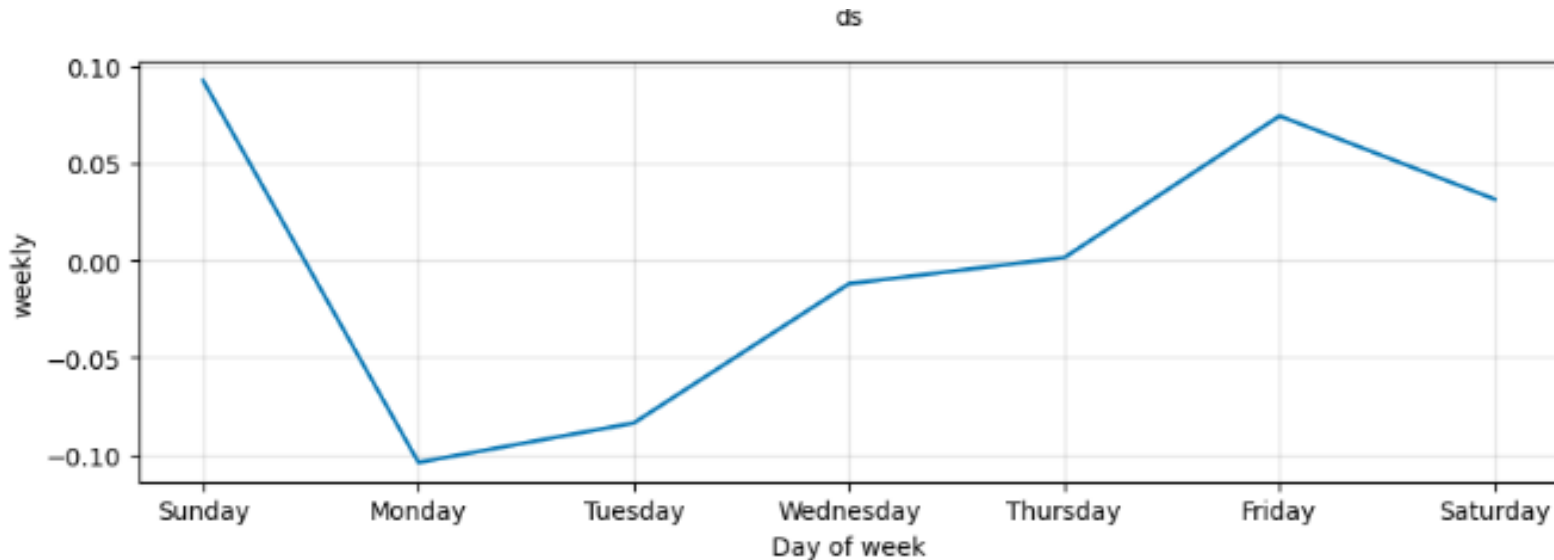
Sales forecast

- The trend is **consistently increasing** over time.
- This indicates a **steady upward growth** in the target variable throughout the forecast period.



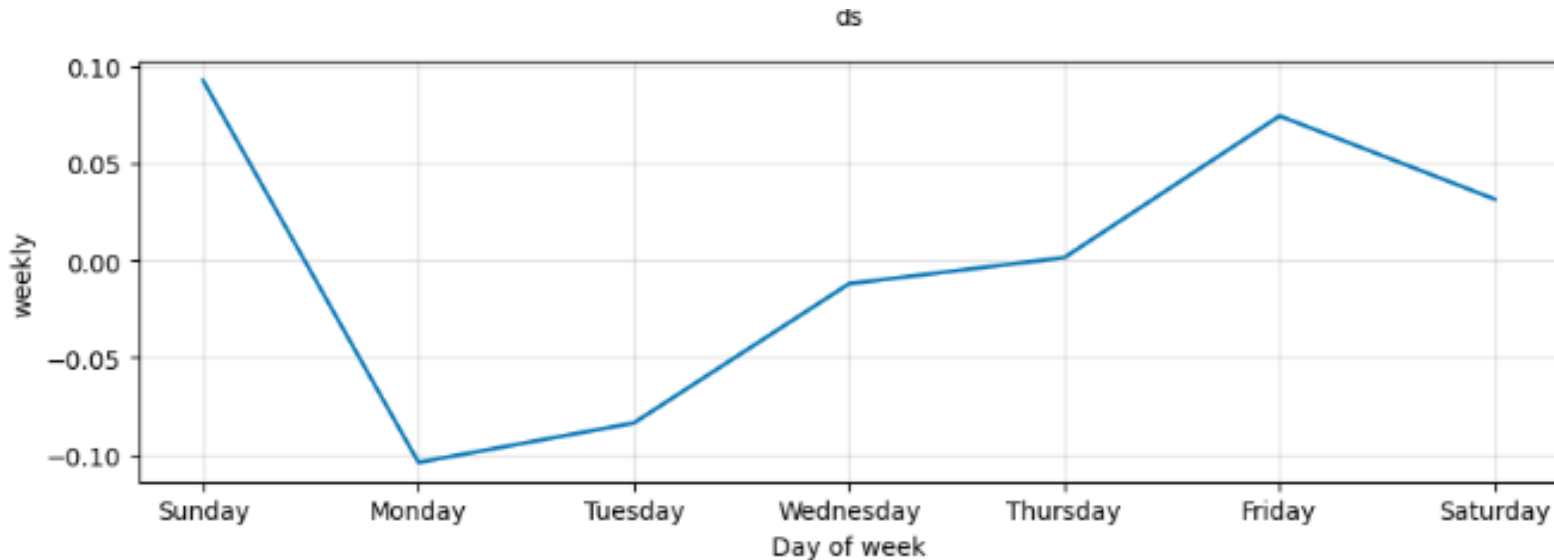
Weekly Seasonality

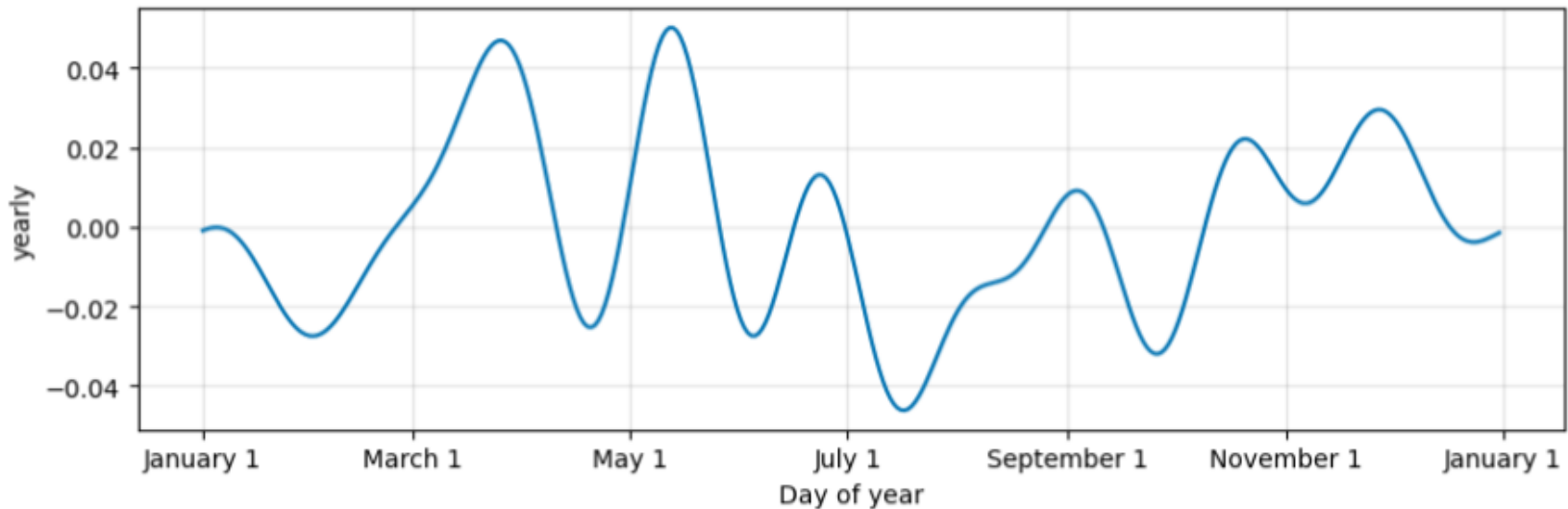
- **Business Strategy:** Focus marketing or promotions on **Sunday and Friday** to maximize returns.
- **Inventory/Staffing:** Prepare for **low activity on Mondays** and scale down operations if needed.
- **Operational Planning:** Plan restocks or internal operations for early weekdays when sales are slower.



Weekly Seasonality

- **Business Strategy:** Focus marketing or promotions on **Sunday and Friday** to maximize returns.
- **Inventory/Staffing:** Prepare for **low activity on Mondays** and scale down operations if needed.
- **Operational Planning:** Plan restocks or internal operations for early weekdays when sales are slower.





Monthly Seasonality

- Peak Periods:** Around **April–May** and **October–November** — best for launching campaigns or increasing stock.
- Low Periods:** Around **late June to mid-July** — potentially due to holidays or seasonal slumps.
- Planning:** Use this seasonal pattern to align **sales, marketing, and inventory** decisions with peak periods.

Conclusion

Linear
Regression
effective

$R^2 = 98.99\%$

No overfitting

Other models
validated

Prophet gave
visual insights